Math 105L **Lecture 9-1**

Differentiating Exponential Functions

Back to Calculus...

- Given a function $f(x)$, its **derivative function** is given by the limit of the difference quotient:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

- To approximate the derivative of a function at a point $x = a$, we plug in a small non-zero value of h into the difference quotient:

$$f'(a) \approx \frac{f(a+h) - f(a)}{h}$$

- **Proportionality Definition:** The quantity A is *directly proportional* to the quantity B if $A = k \cdot B$ for some constant k.

Preliminaries: Playing with Derivatives via Numerical Exploration

Let $f(x) = 2^x$ and $g(x) = 3^x$. Using a small step $h = 0.01$, we compute the ratio of the derivative to the function value:

- **Sample calculation at $x = -4$ for $f(x) = 2^x$:**

$$f(-4) = 2^{-4} = 0.0625, \quad f'(-4) \approx \frac{2^{-4+0.01} - 2^{-4}}{0.01} \approx 0.0435 \implies \frac{f'(-4)}{f(-4)} \approx \frac{0.0435}{0.0625} \approx 0.696$$

- Across all x, the ratio $\frac{f'(x)}{f(x)}$ is constantly ≈ 0.693, which yields:

$$\frac{\frac{d}{dx}2^x}{2^x} \approx 0.693 \implies \mathbf{\frac{d}{dx}2^x \approx 0.693 \cdot 2^x}$$

- Similarly for $g(x) = 3^x$, the ratio $\frac{g'(x)}{g(x)}$ is constantly ≈ 1.0987 (or 1.105 for $h = 0.01$):

$$\mathbf{\frac{d}{dx}3^x \approx 1.0987 \cdot 3^x}$$

- **Key Observation:** In both cases, the derivative of an exponential function is **directly proportional to the function itself!**

Examining the Evidence: Why Does This Happen?

Two fundamental questions arise:

1. *Why is the derivative of an exponential function directly proportional to the function itself?*
2. *What is the exact constant of proportionality?*

Let's compute $\frac{d}{dx}2^x$ directly using the limit definition of the derivative:

$$\begin{aligned}\frac{d}{dx}2^x &= \lim_{h \to 0} \frac{2^{x+h} - 2^x}{h} = \lim_{h \to 0} \frac{2^x \cdot 2^h - 2^x}{h} \\ &= \lim_{h \to 0} \frac{2^x(2^h - 1)}{h} = 2^x \cdot \lim_{h \to 0} \frac{2^h - 1}{h}\end{aligned}$$

Because 2^x has no h dependency, it pulls completely out of the limit!

- Estimating the remaining limit by plugging in a small value $h = 0.001$:

$$\lim_{h \to 0} \frac{2^h - 1}{h} \approx \frac{2^{0.001} - 1}{0.001} \approx 0.69315$$

- **General Base a:** Repeating for any base $a > 0$:

$$\frac{d}{dx}a^x = \lim_{h \to 0} \frac{a^{x+h} - a^x}{h} = a^x \cdot \lim_{h \to 0} \frac{a^h - 1}{h}$$

A Pretty (and Important!) Derivative & The Natural Base e

General Proportionality Law for Exponential Derivatives:

$$\frac{d}{dx}a^x = k \cdot a^x \quad \text{where} \quad k = \lim_{h\to 0}\frac{a^h - 1}{h}$$

Finding the Special Base Where $k = 1$

Suppose we could find a special base a such that the constant of proportionality is exactly 1 ($k = 1$). For that magic base a, we would have:

$$\mathbf{\frac{d}{dx}a^x = a^x}$$

- When $a = 2 \implies k \approx 0.693 < 1$.
- When $a = 3 \implies k \approx 1.099 > 1$.
- By the Intermediate Value Property, such an a must lie between 2 and 3:
  - Try $a = 2.5 \implies \frac{2.5^{0.001}-1}{0.001} \approx 0.9166$.
  - Try $a = 2.718 \implies \frac{2.718^{0.001}-1}{0.001} \approx 0.99989 \approx 1$.
- **Analytical Derivation of the Base e:**

$$\frac{a^h - 1}{h} \approx 1 \implies a^h - 1 \approx h \implies a^h \approx 1 + h \implies a \approx (1+h)^{1/h}$$

Taking the limit as $h \to 0$, and setting $n = \frac{1}{h}$ (so $h = \frac{1}{n}$ as $n \to \infty$):

$$a = \lim_{h\to 0}(1+h)^{1/h} = \lim_{n\to\infty}\left(1 + \frac{1}{n}\right)^n = \mathbf{e} \approx 2.7182818\ldots$$

Derivative of e^x:

$$\mathbf{\frac{d}{dx}e^x = e^x}$$

The natural exponential function is its own derivative! At every point (x, e^x), the slope of the tangent line equals the y-value.

Finding the General Formula: Derivative of a^x via Chain Rule

- **Preliminary Chain Rule Question:** What is $\frac{d}{dx}e^{kx}$?

$$\text{inner: } u = kx \implies u' = k, \quad \text{outer: } e^u \implies (e^u)' = e^u$$

$$\frac{d}{dx}e^{kx} = e^{kx} \cdot k = \mathbf{ke^{kx}}$$

- **Rewriting Base a into Base e:** Recall from algebra that $a = e^{\ln a}$. Therefore:

$$a^x = (e^{\ln a})^x = e^{(\ln a)x} \quad (\text{here } k = \ln a)$$

- **Applying the Chain Rule:**

$$\frac{d}{dx}a^x = \frac{d}{dx}e^{(\ln a)x} = (\ln a)\cdot e^{(\ln a)x} = \mathbf{(\ln a)a^x}$$

General Exponential Derivative Rule:

$$\mathbf{\frac{d}{dx}a^x = (\ln a)\cdot a^x}$$

Examples from Class:

$$\frac{d}{dx}2^x = (\ln 2)\cdot 2^x \approx 0.69315 \cdot 2^x, \qquad \frac{d}{dx}3^x = (\ln 3)\cdot 3^x \approx 1.0986 \cdot 3^x$$